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Part 1: Scenarios

1: Climbing to an altitude

The aircraft increases thrust and lift and angles upwards until a certain vertical height is reached, then levels off and retains a constant altitude.

2: Landing

The aircraft decreases thrust and lift and angles downwards to make contact with the ground and slowing down to a complete stop.

3: Aircraft speeds up to a cruising airspeed

The aircraft increases thrust until a certain relative speed is reached and maintains that velocity.

4: Starting up the aircraft

The operator initiates the steps to fire up the flight systems while the aircraft sits in place on the ground.

5: Taking off on the runway

The aircraft increases thrust to gain an increase in ground speed while increasing lift to approach take-off conditions.

Part 2: User Stories

1.1: Generating lift

As an operator I want to increase lift on the aircraft so that I can gain altitude.

2.1: Engaging the landing gear

As an operator I want to lower the landing gear so that I can safely make contact with the ground.

3.1: Setting the cruising speed

As an operator I want to set an airspeed value so that the aircraft can maintain a steady cruising speed.

4.1: Firing up the engines

As an operator I want to initiate the engines to start so that I can prepare for take off.

5.1: Moving down the runway

As an operator I want to accelerate the aircraft so that I can approach a take off airspeed.

Part 3: Questions

1.1.1: How does the aircraft generate lift?

The aircraft must alter its aerodynamics to dynamically generate lift.

1.1.2: How does the operator tell the aircraft to generate lift?

The operator must be able to control the lift of the aircraft.

2.1.1: How does the landing gear extend?

The landing gear will have an extended and retracted states.

2.1.2: How does the operator engage the landing gear?

The operator should be able to change the state of the landing gear.

3.1.1: How does the operator set the cruising speed?

The operator sets a cruising speed that changes based on conditions.

3.1.2: How does the aircraft know its air speed?

The aircraft needs to sense its airspeed to set the cruise control

4.1.1: What tells the engines to start up?

The engines should be controlled from the cockpit.

4.1.2: What indicates to the operator that the engines are on?

The operator should have data about the current state of the engines.

5.1.1: When does the aircraft start to move?

The aircraft should only move when everything is set to go.

5.1.2: Who signals the aircraft to move?

The aircraft needs a final go ahead signal.

Part 4: Requirements

1.1.1.1: Aircraft must generate lift

Aircrafts are supposed to fly.

1.1.1.2: Aircraft shall control how much lift it generates

Aircraft lift must be variable to fly

1.1.2.1: Operator must control how much lift is generated

Operator must be able to operate the aircraft.

1.1.2.2: Operator must know how much lift is being generated

Operator must be able to operate the aircraft.

2.1.1.1: Landing gear must extend

Extended landing gear allows the aircraft to land

2.1.1.2: Landing gear must retract

Retracted landing gear keeps it safe in flight and helps aerodynamics.

2.1.2.1: Operator must have a landing gear switch

Operator must be able to control the landing gear

2.1.2.2: Operator must have a landing gear indicator

Operator can't physically see the landing gear while in flight

3.1.1.1: Operator must be able to manually set the cruising speed

Operator should always have an option for manual control

3.1.1.2: Operator must be able to cancel a cruising speed

Operator should be able to control the cruise control

3.1.2.1: The aircraft must always know its current speed

Speed represents an important state of the aircraft

3.1.2.2: The aircraft must communicate its current speed to the operator

The operator should always know the state of the aircraft.

4.1.1.1: The engines should be able to turn on manually

The operator should have manual control

4.1.1.2: The engines should be able to turn on automatically

In case the engine fails midflight, the aircraft should try to re-fire them

4.1.2.1: There must be an engine indicator light in the cockpit

The operator should know the state of the engine

4.1.2.2: There must be an engine indicator sound in the cockpit

The operator should know the state of the engine if he doesn't see the light

5.1.1.1: There must be a green light in the cockpit

The operator should know when everything is set to go

5.1.1.2: There must be communication between the operator and flight control

The operator must get the greenlight from flight control

5.1.2.1: There must be a radio

The operator must be able to talk to flight control

5.1.2.2: There must be a speaker

The operator must be able to hear flight control

Part 5: Specifications

1.1.1.1.1: The aircraft must generate positive lift

Positive lift allows for being airborne

1.1.1.1.2: The aircraft must maintain neutral lift

Neutral lift allows for a steady altitude

1.1.1.2.1: The aircraft must have a lift control in the cockpit

Operator must be able to control lift

1.1.1.2.2: The aircraft must be able to automatically control lift

System must be able to correct in a bad situation

1.1.2.1.1: The cockpit lift control must be accessible by the operator at all times

The operator must have physical control

1.1.2.1.2: The cockpit lift control must override the automatic control

A human should always have override

1.1.2.2.1: There must be a dial showing current lift in the cockpit

The operator should always know how much lift is being generated

1.1.2.2.2: The dial should be immediately viewable by the operator

The operator should see the dial

2.1.1.1.1: Landing gear must stick straight down

In order to absorb the shock of impact

2.1.1.1.2: Landing gear must lock into extended position

So that it doesn't collapse on impact

2.1.1.2.1: Landing gear must be able to retract into the body of the aircraft

So that it doesn't get damaged in flight

2.1.1.2.2: Landing gear must lock into retracted position

So that it doesn't flop around while the aircraft is flying.

2.1.2.1.1: There must be a switch in the cockpit labeled landing gear

So that the operator knows what switch it is

2.1.2.1.2: The switch must be reached by the operator

So that the operator can control the landing gear from his seat

2.1.2.2.1: There must be a landing gear status light in the cockpit

So that the operator knows the present state of the landing gear

2.1.2.2.2: The status light must have states for extended and retracted

So that the operator can immediately determine the state of the landing gear

3.1.1.1.1: There must be a lever in the cockpit to set cruising speed

So that the operator can manually adjust the cruise control

3.1.1.1.2: The cockpit lever must have speeds written on it

So that the operator can tell what speed he set it to

3.1.1.2.1: There must be a button in the cockpit to cancel the cruise control

So that the operator can immediately regain control of the engines

3.1.1.2.2: There must be a status light to tell if cruise control is engaged

So that the operator can tell when cruise is manual or automatic

3.1.2.1.1: There must be a speed sensor on the exterior of the aircraft

So that the aircraft can continuously measure its airspeed

3.1.2.1.2: The speed sensor must measure relative speed

Airspeed is relative speed between ground speed and speed of the air

3.1.2.2.1: The speed sensor must send constant signals to the cockpit

The operator should always know the current speed

3.1.2.2.2: There must be a speed gauge in the cockpit

The operator should be able to see the current speed

4.1.1.1.1: There must be a button in the cockpit to turn on the engine

The operator should be able to turn on the engine from the cockpit

4.1.1.1.2: There must be a status light in the cockpit that shows if the engine is on

The operator should know if the engines are on

4.1.1.2.1: The computer must be able to turn on the engines

In case of emergency the computer should try to re-engage the engines

4.1.1.2.2: The cockpit must have an indicator that the computer is engaging the engines

So that the operator knows when the computer is in control

4.1.2.1.1: The engine indicator light must be viewable from the operator seat

So that the operator can always see the state of the engine.

4.1.2.1.2: The indicator light must be green

Green means go

4.1.2.2.1: The engine indicator sound must be audible from anywhere in the cockpit

If the operator is distracted he must have an audible signal

4.1.2.2.2: The engine indicator sound must represent engine off and engine on states

The operator should know the difference between the sounds

5.1.1.1.1: The all systems go light must be green when the plane is safe to take off

A final visual indicator that everything is ready

5.1.1.1.2: The all systems go light must be visible by the operator from the seat

The operator must be able to see the light

5.1.1.2.1: There must be a microphone near all operators in the cockpit

All pilots must be able to communicate with flight control

5.1.1.2.2: The microphone must always be recorded and accessible by the flight control team

Flight control must be able to monitor all flight activity

5.1.2.1.1: There must be a backup radio in the cockpit

Must be a backup for pilots to communicate with other people in their airspace

5.1.2.1.2: The backup radio must be able to be manually tuned

For emergency situations

5.1.2.2.1: There must be a speaker in the cockpit for flight control to directly access

An override speaker for flight control to talk directly to everyone in the cockpit in an emergency

5.1.2.2.2: The speaker in the cockpit must be loud enough to hear in an emergency

For emergency situations

Part 6: Requirement Verification

1.1.1.1.A.1: Is positive lift generated?

1.1.1.2.A.1: Can aircraft control lift?

1.1.2.1.A.1: Can operator control lift?

1.1.2.2.A.1: Can operator determine current amount of lift?

2.1.1.1.A.1: Does landing gear extend?

2.1.1.2.A.1: Does landing gear retract?

2.1.2.1.A.1: Can the operator press landing gear switch?

2.1.2.2.A.1: Can operator determine landing gear position?

3.1.1.1.A.1: Can operator set a cruising speed?

3.1.1.2.A.1: Can operator cancel the current cruising speed?

3.1.2.1.A.1: Can the aircraft determine current speed?

3.1.2.2.A.1: Can operator see the current speed?

4.1.1.1.A.1: Can operator manually turn on engines?

4.1.1.2.A.1: Can engines turn on automatically?

4.1.2.1.A.1: Does engine indicator light turn on?

4.1.2.2.A.1: Does engine indicator sound chime?

5.1.1.1.A.1: Does the all systems go light turn on?

5.1.1.2.A.1: Can operator speak to flight control?

5.1.2.1.A.1: Is there a backup radio present in cockpit?

5.1.2.2.A.1: Is there an emergency loudspeaker in cockpit?

Part 7: Specification Verification

1.1.1.1.1.1: Does aircraft generate enough lift to fly?

1.1.1.1.2.1: Can aircraft maintain steady lift?

1.1.1.2.1.1: Can pilot control lift from the cockpit?

1.1.1.2.2.1: Can the aircraft control lift without pilot input?

1.1.2.1.1.1: Can the pilot always reach the lift control feature?

1.1.2.1.2.1: Can the pilot override the automated lift control?

1.1.2.2.1.1: Does the cockpit dial always show current amount of lift?

1.1.2.2.2.1: Can the pilot always see the lift dial in the cockpit?

2.1.1.1.1.1: Does the landing gear stick straight down?

2.1.1.1.2.1: Does the landing gear lock in the down position?

2.1.1.2.1.1: Does the landing gear fully retract?

2.1.1.2.2.1: Does the landing gear lock in the up position?

2.1.2.1.1.1: Does the landing gear switch operate the landing gear?

2.1.2.1.2.1: Can the pilot reach the landing gear switch?

2.1.2.2.1.1: Is there a landing gear status light?

2.1.2.2.2.1: Does the landing gear status light indicate landing gear position?

3.1.1.1.1.1: Is there a cruising speed lever?

3.1.1.1.2.1: Does the lever indicate what cruising speed it is set to?

3.1.1.2.1.1: Is there a button to cancel the cruise control?

3.1.1.2.2.1: Is there a status light for the cruise control?

3.1.2.1.1.1: There is a pitot tube on the exterior of the aircraft?

3.1.2.1.2.1: Does the pitot tube measure airspeed accurately?

3.1.2.2.1.1: Does the cockpit register the readings from the pitot tube?

3.1.2.2.2.1: Does the speedometer in the cockpit reflect the airspeed?

4.1.1.1.1.1: Is there a button to turn on the engines?

4.1.1.1.2.1: Does the status light show the current state of the engines?

4.1.1.2.1.1: Can the computer turn on the engines?

4.1.1.2.2.1: Does the computer display that it is currently engaging the engines?

4.1.2.1.1.1: Can the pilot see the status lights?

4.1.2.1.2.1: Is the light green?

4.1.2.2.1.1: Is there an audible sound indicating engines are turning on?

4.1.2.2.2.1: Are the audible sounds distinguishable from one another?

5.1.1.1.1.1: Does the all systems go light turn on when the aircraft is ready to depart?

5.1.1.1.2.1: Can the pilot see the all systems go light?

5.1.1.2.1.1: Does the microphone transmit the pilot's voices?

5.1.1.2.2.1: Can the flight control team access the microphone in the cockpit?

5.1.2.1.1.1: Is there a backup radio in the cockpit?

5.1.2.1.2.1: Can the backup radio be manually operated?

5.1.2.2.1.1: Can the flight control team access the speakers in the cockpit?

5.1.2.2.2.1: Can the pilots hear the emergency speakers?